# Counting
# CS1010 Discrete Mathematics for Computer Science

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# Generalized Permutations and Combinations

# Permutations with Repetition

Show that the number of $r$-permutations of a set of $n$ objects with repetition allowed is $n^r$.

**Proof:** There are $n$ ways to select an element of the set for each of the $r$ positions in the $r$-permutation when repetition is allowed. Hence, by the product rule there are $n^r$ ways.

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# Combinations with Repetition

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Vanilla, Chocolate, Strawberry, Mango, Mint.

You would like to buy an ice cream with **8 scoops**. A flavor may be chosen multiple times. How many different ice creams are possible?

Example (Put the scoops into the corresponding flavor box.):

| 3 | 2 | 0 | 1 | 2 |
|---|---|---|---|---|
| V | C | S | M | Mt |

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# Combinations with Repetition (Cont.)

$$\boxed{\text{Vanilla} = 3,\ \text{Chocolate} = 2,\ \text{Strawberry} = 0,\ \text{Mango} = 1,\ \text{Mint} = 2},$$

Represent each scoop by a $\star$ and separate flavors using bars.

$$\star\star\star|\ \star\star|\ |\ \star\ |\ \star\star$$

$\mathcal{A}$. There are 8 stars and 4 bars. Hence, every ice cream corresponds to an arrangement of $8 + 4 = 12$ symbols. Therefore,

$$\binom{12}{4} = \binom{12}{8} = 495.$$

# Combinations with Repetition

## General Rule

The number of ways to choose $n$ objects from $k$ types, where repetition is allowed:

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}.$$

- Suppose the numbers of selected objects of each type are $x_1, x_2, \ldots, x_k$ :

$$x_1 + x_2 + \cdots + x_k = n, \qquad x_i \geq 0.$$

- Each solution can be represented by: $\underbrace{\star \cdots \star}_{x_1} \mid \underbrace{\star \cdots \star}_{x_2} \mid \cdots \mid \underbrace{\star \cdots \star}_{x_k}$.
- Thus, there are $n$ stars and $k-1$ bars and every selection corresponds to an arrangement of $n+k-1$ symbols.
- Choosing the positions of the $n$ stars (or equivalently the $k-1$ bars) gives

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}.$$

# Examples of Combinations with Repetition

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$\mathcal{Q}$. A bakery sells 6 varieties of muffins. How many different ways can a customer buy 7 muffins?

$\mathcal{A}$. Here, $n = 7$ and $k = 6$. Therefore,

$$\binom{7+6-1}{7} = \binom{12}{7} = \binom{12}{5} = 792.$$



# Summary of Formulas

| Type | Repetition Allowed? | Formula |
|---|---|---|
| $k$-permutations | No | $P(n,k) = \frac{n!}{(n-k)!}$ |
| $k$-combinations | No | $C(n,k) = \frac{n!}{(n-k)!k!}$ |
| $k$-permutations | Yes | $n^k$ |
| $k$-combinations | Yes | $C(n+k-1,n) = \frac{(n+k-1)!}{n!(k-1)!}$ |

# Permutations with Indistinguishable Objects

**Theorem**

The number of different permutations of $n$ objects, where there are $n_1$ indistinguishable objects of **type 1**, $n_2$ indistinguishable objects of **type 2**, ..., and $n_k$ indistinguishable objects of **type** $k$, is:

$$\frac{n!}{n_1!n_2!\dots n_k!}$$

**Proof:** By the product rule, the total number of permutations is:

$$C(n, n_1)C(n - n_1, n_2)\dots C(n - n_1 - n_2 - \dots - n_{k-1}, n_k)$$

Manipulating this product algebraically yields the desired result:

$$\frac{n!}{n_1!(n - n_1)!} \cdot \frac{(n - n_1)!}{n_2!(n - n_1 - n_2)!} \cdots = \frac{n!}{n_1!n_2!\dots n_k!}$$

# Permutation of Multisets Example

$\mathcal{Q}$. How many distinct arrangements of the letters in the word `MISSISSIPPI` are possible?

$\mathcal{A}$. The word has 11 letters with repetitions:

| Letter | $M$ | $I$ | $S$ | $P$ |
|---|---|---|---|---|
| Count | 1 | 4 | 4 | 2 |

Hence, the number of distinct arrangements is

$$\frac{11!}{4!\,4!\,2!} = 34{,}650.$$

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Q. A row contains 10 balls: 4 are red, 3 are blue, 2 are green, and 1 is yellow. How many distinct arrangements of the balls are possible?

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$\mathcal{Q}$. A row contains 10 balls: 4 are red, 3 are blue, 2 are green, and 1 is yellow. How many distinct arrangements of the balls are possible?

$\mathcal{A}$. Since balls of the same color are identical, the number of distinct arrangements is

$$\frac{10!}{4!\,3!\,2!} = 12{,}600.$$

# Distributing Objects into Boxes

# Problem Statements

Many counting problems can be solved by counting the ways objects can be placed in boxes.

- The objects may be either
  - **Distinguishable:** different from each other (may be colored, numbered, or otherwise),
  - **Indistinguishable:** identical
- The boxes may be either
  - **Distinguishable:** the boxes may be labeled
  - **Indistinguishable:** the boxes are identical



# Distributing $n$ Objects into $k$ Boxes

| Objects | Boxes | Number of Distributions |
|---|---|---|
| Distinguishable | Distinguishable | $\frac{n!}{n_1!n_2!\cdots n_k!}$ |
| Indistinguishable | Distinguishable | $\binom{n+k-1}{n} = \binom{n+k-1}{k-1}$ |
| Distinguishable | Indistinguishable | Stirling numbers of the second kind $\sum_{j=1}^{k} S(n,j)$ |
| Indistinguishable | Indistinguishable | Integer partitions $p_k(n)$ |

**Only the first two cases have simple closed-form formulas.**

# Q&A: Counting with Repetition

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$\mathcal{A}$. $k = 2$ distinguishable boxes and $n = 8$ indistinguishable objects. So

$$\binom{2+8-1}{8} = \binom{9}{8} = \binom{9}{1} = 9.$$

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$\mathcal{Q}$. How many solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 21$, where the $x_i$ are nonnegative integers satisfying $0 \leq x_1 \leq 3, 1 \leq x_2 < 4, x_3 \geq 15$?

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$\mathcal{A}$. Set $y_2 = x_2 - 1, y_3 = x_3 - 15$. Then $x_1 + y_2 + y_3 + x_4 + x_5 = 5$, with $0 \leq x_1 \leq 3, 0 \leq y_2 \leq 2$.

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$\mathcal{A}$. Set $y_2 = x_2 - 1, y_3 = x_3 - 15$. Then $x_1 + y_2 + y_3 + x_4 + x_5 = 5$, with $0 \leq x_1 \leq 3, 0 \leq y_2 \leq 2$.

- Without the upper bounds, there are $\binom{9}{4}$ solutions. Subtract solutions with $x_1 \geq 4$ and with $y_2 \geq 3$:

$$\binom{9}{4} - \binom{5}{4} - \binom{6}{4} = 126 - 5 - 15 = 106.$$

# Q&A: Multinomial Coefficients

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2 zeros, 3 ones, 5 twos.

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$\mathcal{Q}$. How many ways are there to distribute 12 indistinguishable balls into six distinguishable bins?

$\mathcal{A}$. Let $x_i$ be the number of balls in bin $i$. Then

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 12, \qquad x_i \geq 0.$$

By the Stars and Bars method,

$$\binom{12+6-1}{6-1} = \binom{17}{5}.$$

# Q&A: Permutations with Repeated Objects

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$\mathcal{Q}$. How many different strings can be made from the letters in ABRACADABRA, using all the letters?

$\mathcal{A}$. There are 11 letters in total. Their multiplicities are

$$A : 5, \qquad B : 2, \qquad R : 2, \qquad C : 1, \qquad D : 1.$$

If all letters were distinct, there would be $11!$ arrangements. Since identical letters are indistinguishable, divide by the factorials of their multiplicities:

$$\frac{11!}{5!\,2!\,2!}.$$

Thus,

$$\boxed{\frac{11!}{5!\,2!\,2!} = 83\,160}.$$

# Q&A: Paths in Four Dimensions

**Q.** How many ways are there to travel in $xyzw$-space from $(0, 0, 0, 0)$ to $(4, 3, 5, 4)$ by taking steps one unit in the positive $x$, positive $y$, positive $z$, or positive $w$ direction?



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$\mathcal{A}$. To reach $(4,3,5,4)$, we must take

$$4 \text{ steps in } x, \quad 3 \text{ in } y, \quad 5 \text{ in } z, \quad 4 \text{ in } w.$$

- Thus, there are $4+3+5+4=16$ steps in total. We only need to choose the positions of each type of step:

$$\frac{16!}{4!\,3!\,5!\,4!} = 3\,632\,428\,800.$$

# Inclusion-Exclusion Principle

# The Principle of Inclusion-Exclusion

- Recall **Subtraction Rule**: for two finite sets $A$ and $B$,

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

- Why do we subtract $|A \cap B|$?
- When we add $|A| + |B|$, every element in $A \cap B$ is counted twice.
- Subtracting $|A \cap B|$ leaves every element counted exactly once.
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# Example: Counting Bit Strings

$\mathcal{Q}$. How many bit strings of length 8 either start with a `1`, or end with the bits `00`?



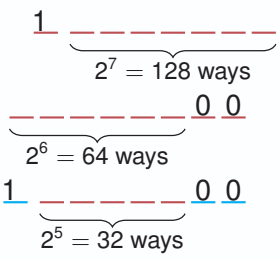


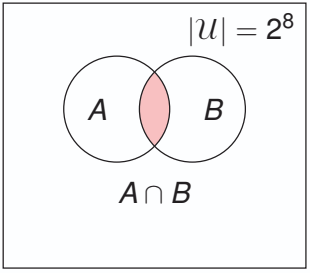


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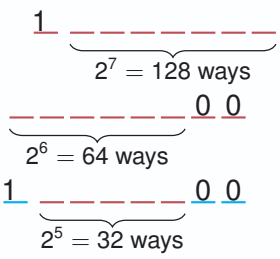


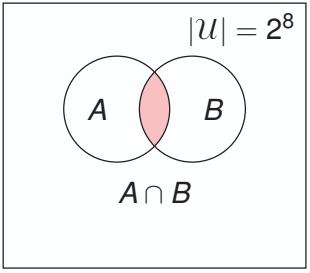




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# Three Finite Sets

- For three finite sets $A$, $B$, and $C$,

$$|A \cup B \cup C| = |A| + |B| + |C| \quad -|A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

- The pattern is:

Add single sets

Subtract pairwise intersections

Add the triple intersection.

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# Three Finite Sets: Example

$\mathcal{Q}$. In a survey of 100 students, 60 like mathematics, 45 like computer science, and 35 like physics. Also, $|M \cap C| = 25, |M \cap P| = 20, |C \cap P| = 15$. If 90 students like at least one of the three subjects, how many students like all three?

$\mathcal{A}$. Let $M, C, P$ denote the sets of students who like mathematics, computer science, and physics. By inclusion-exclusion,

$$|M \cup C \cup P| = 90 = 60 + 45 + 35 - 25 - 20 - 15 + |M \cap C \cap P|.$$

- Hence, $|M \cap C \cap P| = 90 - 60 - 45 - 35 + 25 + 20 + 15 = 10$.

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# Principle of Inclusion-Exclusion

- Let $A_1, A_2, \ldots, A_n$ be finite sets.
- Then

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n| .$$

- Equivalently,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{\emptyset \neq I \subseteq \{1,\ldots,n\}} (-1)^{|I|+1} \left|\bigcap_{i \in I} A_i\right| .$$

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# Proof of Inclusion-Exclusion

- Consider an element $a$ that belongs to exactly $r$ of the sets $A_1, \ldots, A_n$.
- In the first summation, we add $\binom{r}{1}$ counts for $a$.
- In the second summation, we subtract $\binom{r}{2}$ counts for $a$.
- In the third summation, we add $\binom{r}{3}$ counts for $a$.
- So on ....
- More generally, in the $m$-fold intersection summation, $a$ is counted $\binom{r}{m}$ times.
- Hence, the total number of times $a$ is counted is

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1}\binom{r}{r}.$$

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# Proof of Inclusion-Exclusion

- From the Binomial Theorem,

$$(1-1)^r = \sum_{j=0}^{r} (-1)^j \binom{r}{j} = 0.$$

- Therefore,

$$1 - \binom{r}{1} + \binom{r}{2} - \cdots + (-1)^r \binom{r}{r} = 0.$$

- Rearranging,

$$\binom{r}{1} - \binom{r}{2} + \cdots + (-1)^{r+1} \binom{r}{r} = 1.$$

- Thus every element of the union is counted exactly once.

Hence the Principle of Inclusion-Exclusion holds.

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Inclusion-Exclusion Principle

# Applications of Inclusion-Exclusion: Counting Onto-Functions

# Counting Onto Functions

- We want to count the number of **onto functions**

$$f : A \longrightarrow B \text{ where } |A| = m, |B| = n, \text{ and } m \geq n.$$

- There are total $n^m$ functions from $A$ to $B$ in total.
- For each $i = 1, 2, \ldots, n$, let $P_i$ be the set of functions for which the $i$-th element of $B$ is **missing from the range**.
- Therefore, $\#\{\text{onto functions}\} = n^m - |P_1 \cup P_2 \cup \cdots \cup P_n|$.
- Suppose $k$ particular elements of $B$ are missing. Then every element of $A$ has only $n - k$ possible images.
- Hence, $\left|P_{i_1} \cap \cdots \cap P_{i_k}\right| = (n-k)^m$.
- There are $\binom{n}{k}$ ways to choose the $k$ particular elements of $B$ that are missing.

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# Counting Onto Functions (cont.)

- By the Principle of Inclusion–Exclusion,

$$|P_1 \cup \cdots \cup P_n| = \binom{n}{1}(n-1)^m - \binom{n}{2}(n-2)^m + \binom{n}{3}(n-3)^m - \cdots$$
$$= \sum_{k=1}^{n} (-1)^{k-1} \binom{n}{k} (n-k)^m.$$

- Hence, the number of onto functions is

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# Onto Functions and Stirling Numbers

- Consider an onto function $|A| = m, |B| = n$. Then for any **onto** function $f$, every element of $B$ has at least one element of $A$ mapping to it.
- The elements of $A$ can therefore be divided into $n$ nonempty groups, where two elements of $A$ are in the same group if they have the same image under $f$.

Onto function $\longleftrightarrow$ partition of $A$ into $n$ nonempty blocks

- The number of ways to partition an $m$-element set into $n$ nonempty indistinguishable blocks is the **Stirling number of the second kind**: $S(m, n)$.
- We now associate the $n$ elements of the codomain with the $n$ blocks.

**There are $n!$ ways to do this.**

- Therefore, the number of onto functions from an $m$-element set to an $n$-element set is

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# Thank You for your attention!

Any questions?